

Climate 5000 Series - Gen 4

9K BTU Single Zone Max Performance System

1-Way Cassette IDU

Engineering
Submittal
Sheet



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Overview and Project Information



Project name

Location

Architect

Engineer

General contractor

Reference #

9K Btu Single Zone Max Performance System - 1-Way Cassette IDU

Indoor part number	8-733-962-666
Indoor model number	BMS500-AAU009-1AHZXD
Outdoor part number	8-733-962-688
Outdoor model number	BMS500-AAS009-1CSXHD

Efficiency Data		AHRI 210/240 2023
SEER2		24.0
EER2		14.6
HSPF2 Region IV		12.1
HSPF2 Region V		9.5
COP(47F)		3.71
COP (5F)		1.83

Performance Data		AHRI 210/240 2023
Cooling	Capacity Range	4000-12900
	Rated Capacity	9000
	Current	4.5
	Input Power	610
Heating	Capacity Range	3800-13900
	Rated Capacity (47F)	10900
	Rated Capacity (5F)	10600
	Current	3.8
	Input Power	860

Electrical Data	
Power supply (V / Ø / Hz)	208-230 / 1 / 60
Min. circuit ampacity indoor/outdoor (A)	3.0 / 15.0
Max. fuse indoor/outdoor (A)	15.0 / 15.0

Compressor Data	
Type:	Rotary
Refrigerant oil /oil charge (ml)	VG74 / 350
Input (W)	885
Rated current (RLA/A)	6.0

Piping	
Refrigerant precharge (oz/kg)	35.27 (1.00)
Length for precharge (ft/m)	24.6 / 7.5
Liquid side (in/mm)	1/4" / Ø6.35
Gas side (in/mm)	3/8" / Ø9.52
Additional Charge of Refrigerant (oz/ft)/(g/m)	0.16 / 15
Max. refrigerant pipe length (ft/m)	82.02 / 25
Max. difference in height (ft/m)	49.21 / 15

Operation Range

ODU	Cooling (°F / °C)	-22~122 / -30~50
	Heating (°F / °C)	-22~75 / -30~24
IDU	Cooling (°F / °C)	60~90 / 16~32
	Heating (°F / °C)	32~86 / 0~30

System Data

Refrigerant type		R454B
Design pressure (PSIG)		550 / 340
Dehumidification (pts/hr)		2.3
IDU	Fan motor model	ZKFN-30-8-43
	Fan motor qty (ea)	1
	Fan motor input (W)	46.0
	Fan speed (RPM)	1024 / 994 / 864
	Air flow, Turbo / H / M / L / Si (CFM)	341.4 / 294.3 / 259.0 / 235.4 / N/A
	Approx. Air Throw Range ft (m)	23.0 (7)
	Noise level, Turbo / H / M / L / Si (dB(A))	N/A / 38 / 36 / 33 / 24
	Net weight (lbs / kg)	45.19 / 20.5
ODU	Gross weight (lbs / kg)	82.45 / 37.4
	Drain pipe diameter (mm)	ODØ25mm
	Fan motor model	ZKFN-34-10-1-3
	Fan motor qty (ea)	1
	Fan motor input (W)	50
	Fan motor speed (RPM)	850 / 450
	Air flow, Max (CFM)	1471.5
	Noise level, Max (dB(A))	54.5
	Net weight (lbs / kg)	72.75 / 33
	Gross weight (lbs / kg)	78.7 / 35.7

Accessories

BMS-WT2-XXD (8733962707)	Wired Wall Thermostat
BMS-DRY-XXD (8733969263)	24V Interface
BMS-PK1-7X1 (8733951010)	25ft Lineset

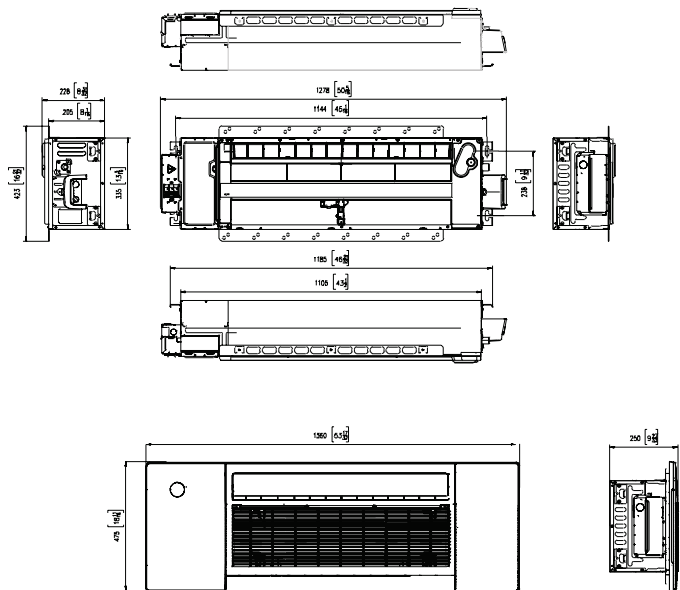
Standard Parts

8-733-969-065	Remote Control
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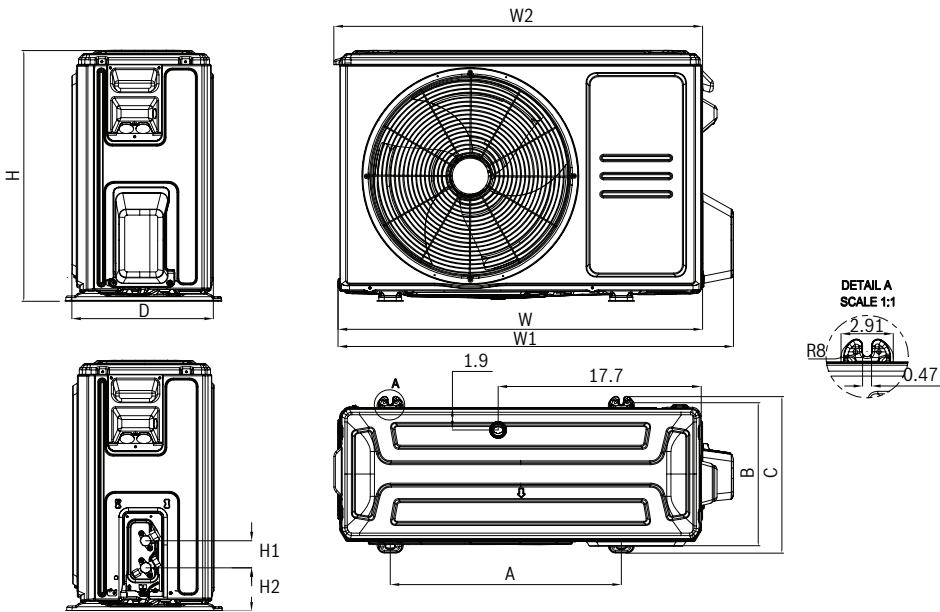


Indoor Unit Dimensions and Connections



Model	W (in)	D (in)	H (in)
BMS500-AAU009-1AHZXD	50.31	13.19	8.98

Outdoor Unit Dimensions and Connections



Model	W (in)	D (in)	H (in)	W1 (in)	W2 (in)	A (in)	B (in)	C (in)	H1 (in)	H2 (in)
BMS500-AAS009-1CSXHD	32	13	21.8	34.4	32.1	20.1	12.5	13.6	2.4	3.8

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Cooling Performance - Indoor Dry Bulb Temperature at 341 CFM																					
ODU DB(°F)	IDU WB (°F)	60.8°F(16.0°C)					64.4°F(18.0°C)					67°F(19.4°C)					71.6°F(22.0°C)				
	IDU DB (°F)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)
-22°F (-30.0°C)	TC	9327	9323	9455	9495	9756	9731	9723	9717	9716	9708	10022	10009	10000	9999	9989	10571	10568	10547	10543	10521
	S/T	0.80	0.91	0.99	0.99	0.99	0.64	0.75	0.84	0.86	0.97	0.55	0.65	0.74	0.75	0.86	0.39	0.47	0.55	0.57	0.67
	PI	0.44	0.44	0.44	0.44	0.44	0.44	0.44	0.44	0.44	0.44	0.44	0.44	0.44	0.44	0.44	0.44	0.44	0.44	0.44	0.44
-13°F (-25.0°C)	TC	10104	10099	10243	10286	10569	10542	10533	10527	10526	10517	10857	10844	10833	10832	10821	11452	11449	11426	11421	11397
	S/T	0.80	0.91	1.00	1.00	1.00	0.65	0.75	0.85	0.86	0.97	0.55	0.65	0.74	0.76	0.86	0.39	0.47	0.56	0.57	0.67
	PI	0.46	0.46	0.46	0.46	0.45	0.45	0.45	0.45	0.45	0.45	0.45	0.45	0.45	0.45	0.45	0.46	0.46	0.46	0.46	0.46
-4°F (-20.0°C)	TC	10881	10876	11031	11077	11382	11353	11344	11336	11335	11326	11692	11678	11667	11665	11654	12333	12330	12305	12300	12274
	S/T	0.80	0.92	1.00	1.00	1.00	0.65	0.76	0.85	0.87	0.98	0.55	0.65	0.74	0.76	0.86	0.39	0.48	0.56	0.57	0.67
	PI	0.47	0.47	0.47	0.47	0.47	0.47	0.47	0.47	0.47	0.47	0.47	0.47	0.47	0.47	0.47	0.47	0.47	0.47	0.47	0.47
0°F (-17.8°C)	TC	11242	11236	11444	11492	11808	11725	11712	11703	11701	11819	12069	12052	12037	12035	12030	12718	12715	12714	12716	12724
	S/T	0.81	0.93	1.00	1.00	1.00	0.66	0.77	0.86	0.88	1.00	0.55	0.66	0.75	0.77	0.87	0.39	0.48	0.57	0.58	0.68
	PI	0.48	0.48	0.48	0.48	0.48	0.48	0.48	0.48	0.48	0.48	0.48	0.48	0.48	0.48	0.48	0.48	0.48	0.48	0.48	0.48
5°F (-15.0°C)	TC	11565	11605	11870	11925	12292	12107	12110	12111	12123	12304	12494	12497	12500	12500	12517	13227	13223	13230	13231	13237
	S/T	0.82	0.94	1.00	1.00	1.00	0.66	0.78	0.87	0.88	1.00	0.56	0.67	0.76	0.78	0.88	0.39	0.48	0.57	0.59	0.69
	PI	0.48	0.48	0.48	0.48	0.48	0.48	0.48	0.48	0.48	0.48	0.49	0.49	0.49	0.49	0.49	0.50	0.50	0.50	0.50	0.50
14°F (-10.0°C)	TC	11086	11080	11306	11359	11711	11636	11626	11618	11616	11725	12035	12019	12007	12005	11993	12797	12793	12768	12764	12736
	S/T	0.81	0.93	1.00	1.00	1.00	0.65	0.76	0.86	0.87	1.00	0.55	0.66	0.75	0.76	0.87	0.39	0.48	0.56	0.57	0.67
	PI	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52
17°F (-8.3°C)	TC	10901	10936	11189	11244	11609	11443	11448	11450	11460	11621	11834	11838	11843	11844	11858	12586	12582	12580	12583	12593
	S/T	0.82	0.93	1.00	1.00	1.00	0.66	0.77	0.87	0.88	1.00	0.55	0.66	0.76	0.77	0.87	0.39	0.48	0.57	0.58	0.68
	PI	0.53	0.53	0.53	0.53	0.53	0.53	0.53	0.53	0.53	0.53	0.53	0.53	0.53	0.53	0.53	0.53	0.53	0.53	0.53	0.53
23°F (-5.0°C)	TC	11292	11341	11634	11692	12077	11864	11868	11869	11885	12090	12274	12278	12281	12281	12301	13050	13046	13056	13057	13065
	S/T	0.83	0.95	1.00	1.00	1.00	0.66	0.78	0.88	0.89	1.00	0.56	0.67	0.76	0.78	0.89	0.39	0.48	0.57	0.59	0.69
	PI	0.53	0.53	0.53	0.53	0.53	0.53	0.53	0.53	0.53	0.53	0.54	0.54	0.54	0.54	0.54	0.55	0.55	0.55	0.55	0.55
35°F (1.7°C)	TC	12188	12188	12260	12294	12691	12758	12757	12758	12758	12792	13157	13156	13157	13157	13158	13898	13894	13894	13893	13894
	S/T	0.78	0.88	0.97	0.99	1.00	0.63	0.73	0.82	0.83	0.93	0.54	0.63	0.72	0.73	0.83	0.39	0.46	0.54	0.56	0.65
	PI	0.49	0.49	0.50	0.50	0.52	0.52	0.52	0.52	0.52	0.53	0.55	0.55	0.55	0.55	0.55	0.59	0.59	0.59	0.59	0.59
41°F (5.0°C)	TC	11951	11952	12047	12130	12489	12507	12507	12508	12508	12552	12898	12897	12897	12898	12898	13621	13618	13617	13616	13618
	S/T	0.78	0.89	0.98	1.00	1.00	0.64	0.74	0.83	0.84	0.94	0.54	0.64	0.72	0.74	0.84	0.39	0.47	0.55	0.56	0.66
	PI	0.49	0.49	0.50	0.50	0.52	0.52	0.52	0.52	0.52	0.53	0.55	0.55	0.55	0.55	0.55	0.60	0.60	0.60	0.60	0.60
47°F (8.3°C)	TC	11715	11722	11879	11933	12286	12258	12257	12258	12257	12314	12638	12637	12638	12638	12638	13346	13343	13341	13341	13342
	S/T	0.79	0.90	1.00	1.00	1.00	0.64	0.75	0.84	0.85	0.96	0.54	0.65	0.73	0.75	0.85	0.39	0.47	0.55	0.57	0.67
	PI	0.49	0.49	0.50	0.50	0.52	0.52	0.52	0.52	0.52	0.53	0.55	0.55	0.55	0.55	0.55	0.60	0.60	0.60	0.60	0.60
50°F (10.0°C)	TC	11430	11745	12141	12211	12685	12705	12682	12165	12204	12697	12546	12563	12569	12570	12691	13411	13447	13481	13485	13502
	S/T	0.90	1.00	1.00	1.00	1.00	0.71	0.85	0.96	0.99	1.00	0.59	0.72	0.83	0.85	0.99	0.40	0.50	0.60	0.62	0.74
	PI	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55
59°F (15.0°C)	TC	11207	11600	12011	12084	12576	11857	11873	12008	12097	12589	12346	12359	12363	12381	12599	13257	13284	13310	13313	13324
	S/T	0.92	1.00	1.00	1.00	1.00	0.73	0.86	1.00	1.00	1.00	0.60	0.73	0.85	0.86	1.00	0.40	0.51	0.61	0.63	0.76
	PI	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57
68°F (20.0°C)	TC	10979	11448	11872	11948	12455	11623	11649	11855	11961	12467	12126	12135	12157	12167	12477	13065	13090	13110	13112	13119
	S/T	0.95	1.00	1.00	1.00	1.00	0.74	0.89	1.00	1.00	1.00	0.61	0.75	0.87	0.89	1.00	0.40	0.51	0.63	0.65	0.78
	PI	0.58	0.58	0.58	0.59	0.59	0.58	0.58	0.59	0.59	0.59	0.59	0.59	0.59	0.59	0.59	0.60	0.60	0.59	0.59	0.59
77°F (25.0°C)	TC	10744	11262	11703	11781	12307	11390	11427	11715	11794	12319	11906	11920	11950	11964	12330	12859	12898	12928	12931	12943
	S/T	0.97	1.00	1.00	1.00	1.00	0.75	0.90	1.00	1.00	1.00	0.61	0.76	0.88	0.90	1.00	0.40	0.52	0.63	0.65	0.79
	PI	0.60	0.60	0.60	0.60	0.61	0.60	0.60	0.60	0.60	0.61	0.60	0.60	0.60	0.60	0.61	0.61	0.61	0.61	0.61	0.61
82°F (27.8°C)	TC	10785	11335	11793	11872	12416	11429	11473	11803	11886	12429	11960	11975	12012	12027	12440	12931	12980	13014	13017	13031
	S/T	0.99	1.00	1.00	1.00	1.00	0.76	0.91	1.00	1.00	1.00	0.62	0.77	0.89	0.92	1.00	0.40	0.52	0.64	0.66	0.80
	PI	0.67	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.69	0.69	0.69	0.69	0.69	0.69
86°F (30.0°C)	TC	11069	11622	12097	12181	12749	11673	11727	12111	12195	12762	12227	12241	12286	12303	12773	13237	13288	13319	13323	13335
	S/T	1.00	1.00	1.00	1.00	1.00	0.77	0.93	1.00	1.00	1.00	0.62	0.78	0.91	0.93	1.00	0.40	0.52	0.64	0.66	0.81
	PI	0.81	0.82	0.82	0.82	0.83	0.82	0.82	0.82	0.82	0.83	0.82	0.82	0.82	0.82	0.83	0.84	0.83	0.83	0.83	0.83
95°F (35.0°C)	TC	11629	11692	12069	12141	12621	12359	12366	1237230												

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Heating Performance - Indoor Dry Bulb Temperature at 341 CFM										
Outdoor DB(°F)	TC Total Heating Capacity(BTU/hr)					PI Power Input(kW)				
	Indoor Conditions(DB °F)					Indoor Conditions(DB °F)				
	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)
-22°F(-30.0°C)	5230	5126	5100	5080	5034	1.35	1.36	1.36	1.36	1.37
-13°F(-25.0°C)	6461	6332	6300	6275	6218	1.41	1.42	1.42	1.42	1.43
-4°F(-20.0°C)	8820	8643	8600	8566	8489	1.51	1.52	1.53	1.53	1.54
0°F(-17.8°C)	9731	9537	9489	9451	9366	1.59	1.60	1.60	1.61	1.61
5°F(-15.0°C)	10871	10653	10600	10558	10463	1.68	1.69	1.70	1.70	1.71
14°F(-10.0°C)	11871	11633	11575	11529	11425	1.71	1.72	1.73	1.73	1.74
17°F(-8.3°C)	12204	11960	11900	11852	11746	1.72	1.73	1.74	1.74	1.75
22°F(-5.6°C)	12694	12440	12378	12328	12218	1.72	1.74	1.74	1.75	1.75
27°F(-2.8°C)	13184	12921	12856	12805	12689	1.73	1.74	1.75	1.75	1.76
32°F(0.0°C)	13674	13401	13334	13281	13161	1.74	1.75	1.75	1.76	1.77
35°F(1.7°C)	13969	13689	13621	13566	13444	1.74	1.76	1.76	1.76	1.77
37°F(2.8°C)	14016	13736	13667	13613	13490	1.67	1.69	1.69	1.69	1.70
42°F(5.6°C)	14136	13853	13784	13728	13605	1.51	1.52	1.52	1.52	1.53
44.6°F(7.0°C)	14183	13900	13830	13775	13651	1.44	1.45	1.45	1.46	1.46
47°F(8.3°C)	14255	13970	13900	13844	13720	1.34	1.35	1.35	1.35	1.36
52°F(11.1°C)	14435	14146	14076	14019	13893	1.18	1.19	1.20	1.20	1.20
57°F(13.9°C)	14615	14323	14251	14194	14067	1.03	1.04	1.04	1.05	1.05
59°F(15.0°C)	14687	14394	14322	14264	14136	0.97	0.98	0.98	0.98	0.99
62°F(16.7°C)	14795	14499	14427	14369	14240	0.88	0.89	0.89	0.89	0.90
64.4°F(18.0°C)	14882	14584	14511	14453	14323	0.81	0.82	0.82	0.82	0.82
69.8°F(21.0°C)	15076	14775	14701	14642	14510	0.65	0.65	0.65	0.65	0.66
75.2°F(24.0°C)	15271	14965	14891	14831	14698	0.48	0.49	0.49	0.49	0.49

Bolded values were tested. All other values are simulated and for reference only.